

Monetization mechanisms in gacha games: The behavioral triad of pricing strategies, pity systems, and belief of luck

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ARTICLE INFO

Keywords:

Virtual economies
Gacha game
Monetization mechanisms
Perceived risk
Pity system
Belief of luck

ABSTRACT

As a dominant monetization mechanism in mobile gaming, gacha systems have raised significant concerns regarding their behavioral impacts due to the probabilistic nature of virtual item acquisition. While previous studies have drawn parallels between gacha engagement and gambling-related disorders, this research adopts a behavioral economics lens to investigate the determinants of user participation in this prevalent virtual economy model. Through a randomized controlled trial ($N = 457$), we systematically examine how pricing strategies (single-pull gacha cost) and pity systems (guaranteed prize mechanisms) interact to shape players' intention to pay. Empirical evidence reveals that both pricing strategies of single gacha and pity systems significantly impact the spending intentions through cognitive reappraisal of perceived risk. Notably, individual differences in belief of luck patterns emerged as critical moderators. This study quantifies the economic interplay between system architecture and superstitious cognition, and provides evidence-based recommendations for gacha mechanisms design and regulatory interventions targeting compulsive spending patterns in digital environments.

1. Introduction

The gacha method is a probabilistic sales approach for virtual items commonly utilized in mobile gaming products. Gacha mechanism originated from the capsule toys initially distributed through vending machines by Japan's Mandai company, known as gacha toys. After 2011, the concept of gacha was integrated into mobile games and has since become an integral component of the industry's business model [1]. Similar to the gacha toy format, users "pull" a gacha by expending a specific amount of in-game currency or exclusive items and subsequently receive a randomly generated bonus item. The rewards obtained from mobile game gacha can vary in terms of type and functionality based on the nature of the game and its content design [2,3].

Rewards obtained from gacha mechanics may alter the appearance of characters and items, providing users with hedonic value through cosmetic enhancements like character skins. Alternatively, these rewards can offer utilitarian benefits by enhancing player capabilities, facilitating gameplay progression, or granting competitive advantages through premium characters and weapon upgrades [13,14,15,7]. These gacha rewards also vary in rarity and value, with rarer and more desirable rewards being less likely to be obtained (Garrett et al., 2022) [16]. In certain gacha games, the game operator regularly updates the

card pool through a "limited time open" format. Failure to participate in the gacha within the limited time frame may result in missing out on obtaining coveted roles. Studies have indicated that the time-limited attribute of gacha will heighten users' perceived stress, thereby stimulating consumption behaviour [17]. The primary motivation for the vast majority of game users to participate in gacha is to obtain the big prize from the prize pool. While users may receive some rewards when they do not win the big prize, these rewards are significantly less rare and valuable than the grand prize [18]. Therefore, this study considers users' participation in gacha and their likelihood of winning as akin to obtaining the grand prize in a gacha pool.

While the gacha mechanism has undoubtedly brought substantial benefits to the gaming market, it is important to acknowledge that for mobile game users, the gacha model can potentially give rise to a myriad of issues. Li, Mills & Nower [19] conducted a study examining the behavior of video game users in relation to purchasing gacha, engaging in video gaming, and participating in gambling. The findings revealed that individuals who made purchases for gacha tended to engage in more frequent and prolonged video gaming sessions, as well as exhibiting higher levels of participation in online gambling activities. These behaviors were associated with an increased likelihood of developing both gambling and video game addiction. Several similar studies have

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Table 1
Study instrumentation of the experiment.

Construct	Item	Measurement	Reference
Perceived risk	PR1	I feel that it is risky to pull the gacha.	Shapiro et al. [4]
	PR2	Given the uncertainty of gacha results, involved in the gacha is risky.	
	PR3	Considering the amount of money associated with purchasing a gacha pull, it is risky.	
	PR4	High nancial risk is involved when purchasing a gacha pull.	
Belief of luck	BL1	I consider myself to be a lucky person.	Chiu & Storm [5]
	BL2	I often feel like it's my lucky day.	
	BL3	I consistently have good luck.	
	BL4	Luck works in my favor.	
Intention to pay	IP1	I intend to obtain the gacha prize for my characters.	Wang et al. [7]
	IP2	My willingness to pay for the gacha is high.	
	IP3	The likelihood that I would engage the gacha is high.	
Hedonic value	HV1	Using the gacha prize is fun.	Wang et al. [7]
	HV2	Using the gacha prize is exciting	
	HV3	The gacha prize make the game enjoyable.	
	HV4	The gacha prize make the game entertaining.	
Utilitarian value	UV1	The gacha prize is helpful for game playing.	Fu [10]
	UV2	The gacha prize is useful for me.	
	UV3	The gacha prize is functional for me.	
Social value	UV4	The gacha prize is practical.	Wang et al. [7]
	SV1	Using the gacha prize would make a good impression on others.	
	SV2	The gacha prize would give its owner social approval.	
	SV3	The gacha prize give me more of a unique identity.	
	SV4	Using the gacha prize would get the attentions of others.	

yielded consistent findings, demonstrating a significant association between paid in gacha and gambling behavior as well as video game overuse [20,21,3]. However, it is crucial to acknowledge the unclear causal direction of this relationship. It remains uncertain whether participation in gacha leads individuals to develop gambling addiction or if gambling prompts increased spending on gacha [22]. Furthermore, certain studies posit that probabilistic sales methods such as gacha may engender cognitive errors among users, including misperceptions of probability, self-rationalization, and operant conditioning, ultimately resulting in irrational consumption behaviors [23,24]. Such irrational consumption behaviors may subsequently give rise to feelings of regret, self-doubt, and other negative emotions among users.

Given the potential adverse impacts of the gacha mode, regulatory authorities in the gaming industry have directed their attention towards scrutinizing the design of game gacha mechanisms. In April 2020, both the U.S. (ESRB) and European (PEGI) video game regulatory committees announced their intention to affix a warning label on games featuring random sales, such as gacha: “In-game purchases (including random items)” [16]. China’s regulations pertaining to the gacha model primarily center around the probability of obtaining desired items, mandating that gacha games disclose the likelihood of obtaining rewards with precision down to a few pulls. Consequently, a “pity system” has emerged within gacha games wherein after a certain number of pulls (N times), players are guaranteed to win [3]. However, unlike fixed probability lotteries like traditional “lotteries,” probabilities within game gachas can be manipulated by operators without transparency for users regarding these alterations.

While there exist regulations and limitations on the gacha

mechanism to a certain extent, game users are unable to accurately discern how operators manipulate and alter the probability of winning. Consequently, it becomes challenging for users to determine the appropriate amount to invest in participating in the gacha as well as ascertain the reasonable price of virtual items [25].

Although researchers and gaming industry professionals have shown increasing interest in gacha game mechanics, current literature lacks empirical evidence on how gacha system design influences user behavior. This study addresses this gap by examining gacha mechanics in mobile games from a product design perspective, specifically investigating their impact on user spending behavior. Furthermore, it evaluates the effectiveness of current regulatory measures. Our findings not only bridge the research gap in understanding the relationship between game design and user behavioral intentions but also provide practical guidance for promoting responsible spending among players.

2. Literature review

2.1. Perceived risk

Perceived risk pertains to the potential for uncertain and unfavorable outcomes perceived by consumers when making purchases of goods or services [26,27,28,29]. [30] proposed a two-factor model of perceived risk, positing that perceived risk emanates from two dimensions: the uncertainty of behavioral outcomes and the adverse consequences resulting from those behaviors. In the gacha game, the probabilistic selling model induces user uncertainty regarding the outcome of their actions, while the financial investment yields unfavorable returns in terms of participation in the gacha. For users of gacha games, the price of a single gacha pull is the most straightforward indicator about the cost of participate in gacha [31]. While there can be significant variation in both the price and the actual number of pulls needed to win, a simple description of probability does not provide game users with a clear understanding of the total investment required. Generally, winning probabilities in gacha games are quite low; for example, in China’s most popular gacha game, the winning rate is typically below 2 %. A lower probability may heighten users’ perceived risk, but the introduction of a pity system serves to mitigate uncertainty surrounding gacha results. Overall, this system provides users with assurance that they will achieve a win within a predetermined number of attempts. In this study, we have inferred that the unit price of gacha and the implementation of a pity system can significantly impact users’ perceived risk and subsequently influence their intention to pay for participation in gacha, even when the probability of winning remains constant. Therefore, we assume that:

H1: The unit price of a gacha game positively affects users’ perceived risk, thereby reducing users’ intention to pay for gacha game.

H2: The pity system of a gacha game negatively affects users’ perceived risk, thereby enhancing users’ intention to pay for gacha game.

2.2. Pity system and over budget spending

Take the popular gacha game “Genshin Impact” in China as an example, when the winning rate is 1.6 % and the pity system times is 90 times. The actual winning rate is 0.6 % until the 73rd pull, after which the probability increases until the 90th pull reaches 100 %. This variable probability and pity system mechanism makes every failed gacha meaningful for the user. The user may be inclined to continue to participate in the gacha pull because continuing to pull after the gacha fails gives the user the opportunity to reduce the losses already incurred [3]. Chen & Fang [32] also obtained a similar result through Markov decision model. In the case of the existence of pity system mechanism, users should either not participate in the gacha, or until the prize is gotten. Based on the aforementioned theoretical analysis, we hypothesize that when the user’s budget is constrained, the user might not be able to obtain the preferred prize within the budgetary range. At such

times, they are prone to continue participating in gacha and give rise to over-budget consumption behavior. Therefore, we assume that:

H3: The pity system of a gacha game will lead to irrational consumption behavior of users with limited budget.

2.3. Belief of luck

Belief of luck is an irrational belief that represents an individual's subjective assessment of their own degree of luck. Some research findings have indicated that luck beliefs exert an influence on individual gambling behavior. For gamblers who perceive themselves as lucky, they are more prone to explain winning in a gamble through internal attribution (that is, winning is regarded as the consequence of personal luck), while attributing losing outcomes to unstable external factors (such as environmental issues or remediable personal flaws) [33,34,35]. Similar studies have also discovered that optimists are more likely than pessimists to sustain positive expectations during gambling and continue gambling after encountering negative gambling outcomes [36]. Other studies have revealed that a considerable proportion of young individuals have higher belief of luck, hold higher expectations of winning in gambling, and are convinced that they can "defeat the system" [37]. As the gacha activity is inherently based on the attribute of probability, this study incorporates luck belief as an individual trait into the model. Users with a stronger belief in luck might think that they have a greater likelihood of winning a gacha game and disregard the risks associated with the gacha game itself. Hence, we assume that:

H4: Belief of luck will weaken the impact of perceived risk on users' intention to pay in gacha game.

To summarize, Fig. 1 illustrates the research model proposed in this study.

3. Methodology

3.1. Procedures and manipulations

An online experiment was employed to validate the theoretical model in this study. We employed a 3 (price for one gacha: ¥6 vs. ¥16 vs. ¥36) * 2 (with pity system vs. no pity system) scenario-based experimental design, whereby participants were randomly assigned to one of six groups. The 3 levels price of single gacha pull employed in the experiment were formulated by referring to the prices of popular gacha games in the Chinese game market. Based on the actual price in the game (¥16), values both higher and lower were respectively adopted. The probability of winning the gacha was consistent in all experimental groups, which is 1.6 %.

Prior to the beginning of the experiment, participants underwent an initial screening process. Those totally have no experience of gacha games been excluded, to guarantee that participants could fully comprehend the relevant content. Those who passed the primary

screening were randomly assigned to an experimental group and initiated the experiment.

Firstly, participants completed a questionnaire to report relevant demographic details, game usage and payment experiences, as well as the participant's level of belief of luck. Subsequently, participants read a scenario description corresponding to the experimental group, which required them to imagine a new gacha prize in a gacha game they had recently utilized. The grand prize was an "advanced character that met the participant's game requirements." Owing to the variations in individual preferences and in-game needs among game users, such differences would result in diverse perceived hedonic and utilitarian values of users for gacha prizes and influence their participation intentions. To prevent such individual differences from influencing the experimental outcomes, the researchers did not present specific information regarding the gacha prize during the experiment but instead asked participants to imagine the prize. The scenario description further provides information regarding the gacha, encompassing the price of a single gacha, the existence of a pity system, the winning probability, and the time limit for the gacha. During the experiment, participants were presented with a simulated gacha animation effect, enabling them to better simulate the gacha experience in a real mobile game (see Appendix A for additional details). Subsequently, the participants' perceived risk of the gacha, willingness to pay for participation, budget for the gacha, and perceived value of the imagined gacha prize were measured through questionnaires. Alternatively, participants were requested to envision whether they would desire to continue participating in the gacha if the expected prize had not been obtained within their budget.

3.2. Measures

All survey items were measured using a 7-point Likert-type scale, ranging from 1 (strongly disagree) to 7 (strongly agree). These measures were derived from established instruments in the literature and are meticulously outlined in the Table 1. The measurement scale of perceived risk is adapted from that of Shapiro et al. [4], which primarily measures the currency risk resulting from consumption behavior ($\alpha_{PR} = 0.897$). The belief of luck scale, adopted from Chiu and Storm (2009)[5], assesses individuals' perception of their own luck through four questions ($\alpha_{BL} = 0.892$). The measurement scale of intention to pay stems from the study of Wang et al. [7] regarding willingness to pay in games ($\alpha_{IP} = 0.937$). The perceived value scale is adapted from Fu [10] and [12] to measure the perceived value of gacha prizes from three aspects: hedonic value ($\alpha_{HV} = 0.871$), utilitarian value ($\alpha_{UV} = 0.733$), and social value ($\alpha_{SV} = 0.867$).

3.3. Participants

We recruited participants and collected data using an online survey platform called Credamo (<https://www.credamo.com>). After the pre-

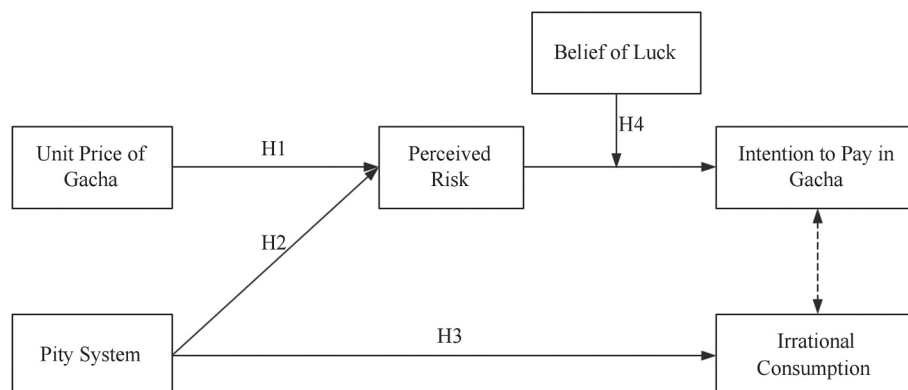


Fig. 1. Xxxx.

screening of participants, a total of 539 participants took part in and completed the experiment. The researchers excluded the data of those participants who failed to provide a specific gacha game name, or had missing data. Eventually, 457 valid data were obtained, and the valid rate was 84.79 %. Table 2 presents the distribution of effective participants among the six experimental groups.

We analyzed the demographic traits of all participants before testing the hypotheses. The gender distribution of the participants in this study is proportionately balanced ($n_{\text{male}} = 198$, 43.327 %; $n_{\text{female}} = 259$, 56.673 %). The majority of participants are concentrated within the age bracket of 21 to 40 ($n_{21-40} = 414$, 90.591 %), which is essentially consistent with the age distribution of users in the game market. From the aspect of weekly game time, it can be discerned that the participants in this study have extensive experience in mobile games, and the majority of them play for between 15 and 35 h per week ($n_{15h-35h}=317$, 69.635 %). Additionally, participants holding a bachelor's degree constituted the vast majority ($n = 357$, 78.12 %). And the participants with an average monthly income ranging from 5001 to 10,000 yuan were the most numerous ($n = 214$, 46.827 %).

In addition, the demographic differences between the groups were tested by chi-square test. The results indicated that there were no significant differences between the groups in terms of gender ($p = 0.083 > 0.05$), age ($p = 0.641 > 0.05$), education level ($p = 0.697 > 0.05$), monthly income ($p = 0.281 > 0.05$), and game usage time ($p = 0.252 > 0.05$), thus they would not affect the experimental results.

4. Results

4.1. Testing hypotheses

Two-factor ANOVA was employed to investigate the differences of perceived risk and intention to pay for gacha among participants in diverse experimental groups (see Table 3a and 3b). The findings indicate that the price of a single gacha exerts a significant influence on the perceived risk ($F = 6.053$, $p = 0.003^{***} < 0.05$) and intention to pay ($F = 21.638$, $p = 0.000^{***} < 0.05$). Likewise, the presence of a pity system mechanism also has a notable impact on the perceived risk ($F = 6.085$, $p = 0.014^{**} < 0.05$) and intention to pay for gacha ($F = 42.881$, $p = 0.000^{***} < 0.05$). No interaction was detected between the pity system mechanism and the price of a single gacha. Employing linear regression, the researchers explored the correlation between perceived risk and users' intention to pay. The findings reveal that perceived risk exerts a significantly negative influence on intention to pay ($\beta = -0.462$, $p = 0.000^{***} < 0.05$).

We further conducted post-hoc multiple comparisons to analyze the specific impacts of the single gacha price and the pity system on perceived risk and intention to pay (see Figure 2 and 3). The outcomes demonstrated that as the price of a single gacha rose, participants' perceived risk of gacha escalated and their intention to pay declined, thereby providing support for H1. Nevertheless, contrary to our hypothesis, the presence of the pity system mechanism augmented the perceived risk of participants and simultaneously enhanced the intention of users to pay. We conjecture that this might be attributed to the high number of must-win times set in the pity system (90 times) and the fact that the must-win times offers more direct information regarding the risk of gacha than the probability. Hence, the must-win times of the pity system strengthened participants' risk perception of gacha.

Table 2
Grouping of participants in the experiment.

With pity system	Price for single gacha(RMB)			Total
	¥6	¥16	¥36	
Yes	$n = 79$	$n = 74$	$n = 74$	$n = 227$
No	$n = 74$	$n = 78$	$n = 78$	$n = 230$
Total	$n = 153$	$n = 152$	$n = 152$	$n = 457$

To test H3, we initially carried out an additional screening of the participant data. Based on the price of a single gacha and the provided probability of winning in the experiment, we computed the mathematical expectation of the funds required to obtain the gacha prize. Subsequently, we regarded participants whose gacha budget was lower than that as users with limited budgets. Among the 457 participants, 411 participants were identified as limited budget users, constituting 89.93 %, which is consistent with the market scenario of free and low-paying users in the mobile game market. Through chi-square comparison, we compared the probability of limited-budget users with and without the pity system to continue gacha after their budget was depleted. The findings indicate that the probability of paying over the budget is significantly higher when a pity system mechanism is present (65.4 % vs. 76.5 %, $X^2 = 6.117$, $p = 0.013^{**} < 0.05$). This provides support for H3.

Subsequently, we explore the moderating influence of luck beliefs on the process by which perceived risk affects the intention to pay. The results are presented in Table 4 below, indicating that an individual's belief in luck mitigates the negative impact of perceived risk on the intention to pay ($\beta = 0.067$, $p = 0.027^{**} < 0.05$). That is, for participants with a stronger belief in luck, their perceived risk of the gacha has a less negative effect on their intention to pay, thereby providing support for H4.

4.2. Supplemental study

In the actual operation of the game product, the probability of winning the gacha is typically set at an extremely low level. Considering the practical implications of the study, we did not incorporate the probability of winning gacha as a crucial feature of the gacha game configuration. Nevertheless, the influence of the probability of winning a gacha on the perceived risk and the intention to pay was taken into account. We carried out a supplementary experiment which aimed to test the effect of winning probability. Based on the probability settings of a popular gacha games in the Chinese game market, we modified the probability of winning the gacha at three levels (0.6 %, 1.6 % & 3 %). Similar to the main experiment in this paper, participants were randomly assigned to an experimental group, read a description of the gacha game, and answered a series of questions.

The supplementary experiment was published through Credamo, an online survey platform, and 363 participants engaged in the experiment. After excluding participants who failed the attention test and did not play any gacha games, valid data were acquired for 338 participants ($n_1 = 113$; $n_2 = 113$; $n_3 = 112$). The chi-square test was employed to compare the age and gender of the participants among the groups, and the results demonstrated that there were no significant disparities in the demographic characteristics of the participants among the groups ($p > 0.05$).

The AVONA was employed to compare the perceived risk and the intention to pay for gacha under diverse winning probability conditions. The findings indicated that there was no significant disparity in the perceived risk ($F = 0.817$, $p = 0.442 > 0.05$) and the intention to pay ($F = 0.593$, $p = 0.553 > 0.05$) among all groups. The outcomes of the supplementary experiment substantiated those minor alterations in the winning probability of gacha at a lower level exerted no significant influence on the participants.

5. Discussion

5.1. Theoretical implications

As an essential virtual item sales model within the contemporary mobile game market, gacha has generated substantial profits for game manufacturers. Nevertheless, there are currently scarce studies on gacha, and the majority of the existing ones concentrate on the association between gacha and gambling as well as other detrimental

Table 3a

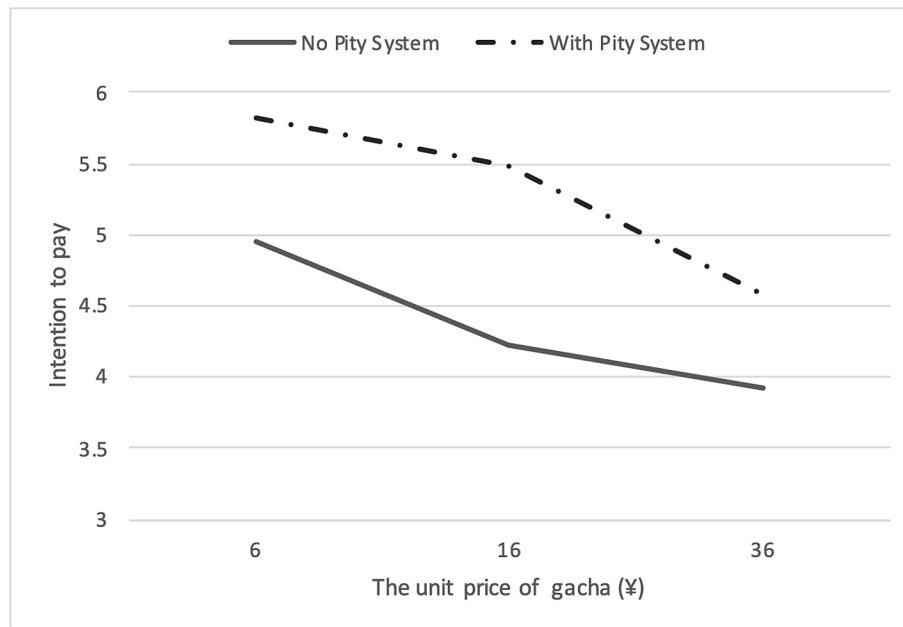
Results of two-way ANOVA of the impact of gacha mechanism on perceived risk.

	Sum of squares	df	Mean square	F	P	R ²	Adjusted R ²
Intercept	11194.267	1	11194.267	5410.779	0.000***	0.041	0.03
Pity system	12.589	1	12.589	6.085	0.014**		
Gacha price	25.044	2	12.522	6.053	0.003***		
Pity system * Gacha price	3.153	2	1.577	0.762	0.467		
Residual	933.066	451	2.069				

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.**Table 3b**

Results of two-way ANOVA of the impact of gacha mechanism on intention to pay.

	Sum of squares	df	Mean square	F	P	R ²	Adjusted R ²
Intercept	10643.526	1	10643.526	4638.233	0.000***	0.168	0.159
Pity system	98.401	1	98.401	42.881	0.000***		
Gacha price	99.306	2	49.653	21.638	0.000***		
Pity system * Gacha price	6.912	2	3.456	1.506	0.223		
Residual	1034.926	451	2.295				

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.**Fig. 2.** Xxxx.

behaviors [21,3]. While these studies provide important insights into the potential harms of gacha, they often overlook how its unique structural elements influence user decision-making. From the perspective of the mechanism design of gacha games, our research demonstrates that the price of single gacha and the pity system mechanism of gacha games can influence users' perceived risk and intention to pay. Consistent with previous research on perceived risk, the higher the price of a single gacha, the greater the perceived risk of users and the lower the intention to pay for it. In particular, we discovered that the existence of a pity system elevates both the perceived risk and the intention to pay for gacha. We contend that in the actual operational process of mobile game products, the probability of winning the grand prize in gacha is always low, and the number of guaranteed winning times set by the pity system is typically high. This direct display of times might offer users more straightforward information compared to the probability of winning and also reduce the difficulty for users to process the information. Hence, the pity system reinforces the user's perceived risk of gacha. Simultaneously, due to the presence of the pity system, the gacha activity is no

longer entirely dependent on probability, and each failed gacha is endowed with a certain significance, thereby enhancing the intention of users to pay. This dual role of the pity system—amplifying risk perception while also enhancing payment intention—contrasts with prior literature that predominantly frames gacha mechanisms purely as risk-inducing or exploitative (e.g., [38,39], thereby offering a more nuanced theoretical account.

In addition, our study reveals the moderating role of users' personal beliefs of luck. We found that the negative effect of perceived risk on payment intention is less pronounced among users with stronger luck beliefs. This finding aligns with and extends the work of Galvin et al. [33], who demonstrated that individual differences in cognitive biases can mitigate the influence of perceived uncertainty in freemium games. Our results further specify how luck beliefs operate within the distinctive context of gacha mechanics—particularly in the presence of explicit safeguard mechanisms such as pity systems. The study advances the understanding of how transparent failure accumulation mechanisms (pity system) create unique psychological value propositions that differ

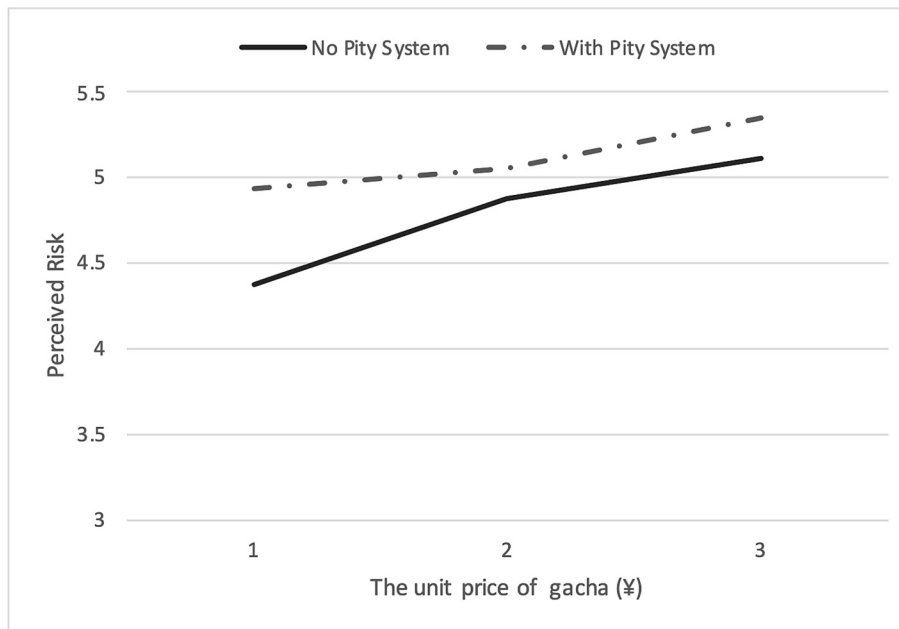


Fig. 3. Xxxx.

Table 4

Test results of the moderating effect of luck beliefs.

	Model 1	Model 2	Model 3
Intercept	6.896** −28.98	4.887** −12.969	6.676** −7.525
Perceived risk	−0.512** (−11.104)	−0.428** (−9.361)	−0.774** (−4.780)
Belief of luck		0.323** −6.688	−0.019 (−0.118)
Perceived risk * Belief of luck			0.067* −2.225
Sample size	457	457	457
R ²	0.213	0.284	0.292
Adjust R ²	0.211	0.281	0.287
F	F (1,455) = 123.295, p = 0.000	F (2,454) = 89.937, p = 0.000	F (3,453) = 62.131, p = 0.000
ΔR ²	0.213	0.071	0.008
ΔF	F (1,455) = 123.295, p = 0.000	F (1,454) = 44.729, p = 0.000	F (1,453) = 4.952, p = 0.027

***p < 0.001, **p < 0.01, *p < 0.05

from those in traditional probability-based systems. These insights underscore the need to expand existing theoretical models of micro-transaction behavior—such as the I-PACE model [40]—by incorporating gacha-specific structural and psychological factors. By integrating these elements, our research offers a more comprehensive theoretical foundation that helps bridge the gap between mechanistic game design features and behavioral outcomes, as highlighted in prior literature.

5.2. Practical implications

In light of the uncertainty of the gacha outcomes, regulatory bodies in various regions have turned increasing attention to the supervision of loot boxes and gacha mechanics. Our research provides empirical evidence that can inform these regulatory efforts and guide ethical game design. We attest that the pity system, a common yet under-researched feature, presents a double-edged sword: it can offer transparency and a form of caution to users, helping them form a more distinct comprehension of the risks, yet it also introduces a novel form of risk that is not fully addressed by current frameworks focused solely on probability disclosure (e.g., the regulations implemented in China).

Our findings demonstrate that the presence of a pity system can inadvertently give rise to users' over-budget consumption by creating a predictable endpoint that justifies continued spending. This is particularly problematic if game operators, who typically only publish an overall win probability, engage in dynamically manipulating the hidden winning odds before a user reaches the pity threshold. This practice, which our study suggests is a potential risk, aligns with industry concerns about "bad luck protection" systems that may not be transparently implemented [39]. Additionally, we validate the role of luck belief in the gacha process. Market feedback suggests that some games might guide user behavior by manipulating this belief; for instance, by temporarily boosting win rates for novices or unlucky players to enhance their confidence and gaming experience. Generally speaking, the manipulation of probabilities by game operators and the concealment of changes in probabilities prompt users to spend more. From a regulatory perspective, move beyond mandating static probability disclosure. Regulations should require the detailed disclosure of probability variation rules (including the specific parameters and triggers of pity systems) and explicitly prohibit the real-time manipulation of odds for individual players. This targeted approach safeguards user rights directly at the mechanistic level our research identified. Simultaneously, game operators can be encouraged to build trust and sustainable monetization, game operators should be encouraged or required to disclose aggregate user gacha data, allowing third parties to audit and verify the accuracy of published probabilities against actual results. This practice would address the core issue of transparency that erodes consumer confidence. For developers, they should avoid exploiting cognitive biases by designing systems that manipulate perceived luck, and instead, focus on creating value propositions that are transparent and fair, thus fostering positive long-term player engagement.

By translating our results into these specific implications, we provide a clear roadmap for stakeholders to address the nuanced risks of gacha mechanics, thereby bridging the gap between academic research, industry practice, and regulatory policy.

5.3. Limitations and future research opportunities

There remain several limitations in our study that offer valuable opportunities for future research. Firstly, in games, users can frequently obtain a certain quantity of free gacha by accomplishing in-game tasks.

The difficulty and frequency with which a user acquires free gacha in a game might also exert an influence on the amount a user pays to participate in a gacha. The effect of free gacha opportunities was not taken into account in this study, as this can vary among different games. Future research could further investigate related issues, which would be conducive to understanding the in-game purchase behavior of low-paying users. Secondly, while this study compared basic demographic differences among participants, it did not deeply explore the role of broader individual user differences. Future research should more comprehensively examine how factors such as age, gender, and gaming experience influence gacha engagement. A particularly critical limitation is the absence of screening for participants' tendencies toward addictive behaviors, including problematic gaming and gambling. This is an important oversight because addictive predispositions could significantly alter risk perception and spending behavior within gacha mechanics. Without controlling for or measuring these tendencies, it is difficult to fully isolate the psychological mechanisms under investigation or to ensure that findings are not confounded by a subset of participants with compulsive consumption patterns. Future studies should therefore incorporate validated scales for measuring gambling and gaming addiction to better understand how gacha mechanics affect vulnerable individuals and to provide a more nuanced analysis of behavioral outcomes. Thirdly, conducted within the specific context of mobile games in China, utilizing a sample of Chinese game users. While this offers valuable cultural specificity, it also limits the generalizability of the findings. Future studies could replicate this research in other cultural and regulatory settings to investigate whether users from different backgrounds exhibit varying levels of willingness to pay for gacha, particularly in regions with stricter loot box regulations or different cultural attitudes toward gambling.

6. Conclusion

This study advances the understanding of gacha game mechanics by demonstrating how key design elements—specifically single-pull price and pity systems—directly influence users' payment intentions through the mediating role of perceived risk. The findings reveal a behavioral paradox wherein the pity system functions both as a risk-signaling mechanism and a spending incentive, potentially resulting in over-budget consumption despite heightened risk perception. Moreover, the study identifies personal luck beliefs as a significant moderating variable that weakens the negative effect of perceived risk on payment intention, indicating that users with stronger superstitions are less susceptible to risk-related deterrents. These insights extend prior research on game monetization (e.g., [21,3]) by (1) empirically validating the psychological pathways through which gacha mechanics affect behavior, (2) illuminating the dual role of guarantee mechanisms—contrasting with earlier works that treated them as purely protective or exploitative [38], and (3) establishing individual differences in luck belief as a critical boundary condition in free-to-play spending models. The study thereby offers refined theoretical perspectives and practical guidance for game designers aiming to align profitability with player welfare and for regulators developing evidence-based policies for virtual item environments.

6.1. Compliance with ethical standards

- Prior to commencing their participation, the participants perused the instructions regarding the relevant content and purpose of the experiment and voluntarily signed an informed consent form.
- During the experiment, it will neither cause harm to the human body nor involve personal sensitive information and commercial interest information.
- Participants have the option to abort and withdraw at any point during the experiment, and their data will not be registered.

Based on this, this study conformed to the fundamental guidelines of the Declaration of Helsinki and the ethical exemption requirements stipulated in the Ethical Review Measures for Life Sciences and Medical Research Involving Human Beings issued by the Chinese government, thereby enabling it to be exempted from ethical approve. Prior to the initiation of the study, the authors' institutions reviewed the content of the study and consented to the exemption from ethical approve.

Ethical approval

The experimental survey of this study was released on an online crowdsourcing platform in China in the form of an online survey.

CRediT authorship contribution statement

Chang Ma: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Conceptualization. **Jingbo Shao:** Supervision, Project administration, Funding acquisition. **Pengyu Li:** Writing – review & editing, Methodology.

Funding

This study was funded by National Natural Science Foundation of China (72272045, 71831005).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: National Natural Science Foundation of China (72272045, 71831005);

Appendix

Simulated gacha interface screenshot.

Data availability

Data will be made available on request.

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